

PAPER_1378 — UQFF Resolution of the Loschmidt Reversibility Paradox (CLOSED — CW/CCW Identity)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Foundational / Thermodynamic (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — $F_{\text{TRZ}} \times \beta_i$ chirality imbalance breaks time-reversal symmetry **Calculator surface:** `calculate_paradox({"paradox": "loschmidt_paradox"})` **Closure helper:** `_l96_uqff_axiom_loschmidt_paradox_closure()`

The Paradox

Loschmidt's Umkehrwand (1876, raised against Boltzmann): the microscopic equations of motion (Newtonian, Hamiltonian, Schrödinger, Dirac) are **time-reversible**. For every forward trajectory there exists a velocity-reversed trajectory of equal weight. The second law's irreversibility ($\Delta S \geq 0$) cannot follow from reversible micro-dynamics without an additional symmetry-breaking ingredient.

The standard answer points to **initial conditions** (low-entropy past hypothesis) — effectively a free parameter. UQFF supplies the symmetry-breaking ingredient structurally.

UQFF Closed Identity

The vacuum has two chirality channels: - **North pole = SCm (CW rotation)** — massless, reactive with trapped Aether - **South pole = UA' (CCW rotation)** — forms when SCm encapsulates free UA

Symmetric under naive parity, but the **time-reversal zone** $F_{\text{TRZ}} = 1/10$ and the dynamic coupling $\beta_i = 0.6029$ produce a **finite chirality imbalance**:

$$\begin{aligned}\text{Arrow asymmetry} &= F_{\text{TRZ}} \times \beta_i = 0.1 \times 0.6029 = 0.0603 \\ \text{Entropy rate} &= K_{\text{MEX}} \times F_{\text{TRZ}} = (25/12) \times 0.1 = 0.2083\end{aligned}$$

A negative-time branch **does exist** (PAPER_597, `_negative_time_dual_existence`) — Loschmidt is right that the equations admit it. But the CW vs CCW pair is asymmetric at the $F_{\text{TRZ}} \times \beta_i$ level, so the forward-time branch carries strictly more weight in the $F_U = 1$ ledger.

$$\begin{aligned}P(\text{forward}) / P(\text{backward}) &= (1 + F_{\text{TRZ}} \times \beta_i) / (1 - F_{\text{TRZ}} \times \beta_i) \\ &= 1.0603 / 0.9397 \\ &= 1.1283\end{aligned}$$

A **~13% directional bias per fundamental tick** — sufficient to exponentially suppress the reverse trajectory over macroscopic times.

Physical Interpretation

- **Loschmidt is correct that the reverse trajectory exists.** UQFF does not deny the `t_neg` branch; PAPER_597 wires it explicitly.
- **The forward / backward asymmetry is structural**, not parametric. $F_{\text{TRZ}} = 1/10 = 1/\text{SO}_5$ and $\beta_i = 0.6029$ are canonical UQFF primitives anchored in PAPER_1203.

- **The $K_{\text{MEX}} \times F_{\text{TRZ}}$ entropy-production rate** = $25/120 = 0.2083$ is the integer-primitive coarse-graining floor that drives $\Delta S \geq 0$ on macroscopic timescales.
- **No initial-conditions assumption required.** The low-entropy past is recovered as a consequence of the $F_{\text{TRZ}} \times \beta_i$ imbalance acting cumulatively backward from any present state.

Live Calculator Output

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "loschmidt_paradox"})["value"]
```

Field	Value
arrow_of_time_asymmetry_via_F_TRZ_x_beta_i	0.0603
entropy_production_rate_via_K_MEX_x_F_TRZ	0.2083
reversibility_broken_via_CW_CCW_chirality_imbalance	True
t_neg_branch_exists_but_F_TRZ_breaks_symmetry	True

C++ Reference Implementation

```
class LoschmidtResolutionUQFF {
public:
    static constexpr double F_TRZ = 0.1;
    static constexpr double BETA_I = 0.6029;
    static constexpr double K_MEX = 25.0 / 12.0;
    static double arrowAsymmetry() {
        return F_TRZ * BETA_I; // 0.0603
    }
    static double entropyProductionRate() {
        return K_MEX * F_TRZ; // 0.2083
    }
    static double forwardBackwardRatio() {
        double e = arrowAsymmetry();
        return (1.0 + e) / (1.0 - e); // 1.1283
    }
};
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard Statistical Mechanics solve the Loschmidt paradox via different methods. SM invokes a **low-entropy past hypothesis** (Boltzmann, Penrose) — an unconstrained initial-condition postulate. UQFF derives:

- **$F_{\text{TRZ}} \times \beta_i = 0.0603$** structural chirality imbalance at every fundamental tick.
- **$K_{\text{MEX}} \times F_{\text{TRZ}} = 0.2083$** entropy-production rate as an integer-primitive identity.
- **The reverse trajectory exists** (PAPER_597 t_neg branch) but is exponentially suppressed by the cumulative $F_{\text{TRZ}} \times \beta_i$ bias.
- **Zero free parameters.** All three primitives — F_{TRZ} , β_i , K_{MEX} — are canonical UQFF locks.

Reference

- UQFF foundational papers: PAPER_597 (negative-time dual existence), PAPER_646 (CW/CCW grinding mechanism), PAPER_1203 ($\beta_i = 0.6029$ canonical), PAPER_062 (DPM vacuum manifold).
- Related closures: `arrow_of_time` (K_MEX entropy gradient), `negative_time_dual_existence` (t_neg branch), `dpm_grinding` (5-step CW $\times$ CCW sequence).
- Closure location: `uqff_pure_calculator.py` $\rightarrow$ `_l96_uqff_axiom_loschmidt_paradox_closure`
- Dispatch: `PARADOX_TO_CLOSURE["loschmidt_paradox"], ["loschmidt"], ["reversibility_paradox"]`

Copyright — Daniel T. Murphy, daniel.murphy00@gmail.com, dated June 16, 2026, location 41.0997° N, 80.6495° W (Youngstown, OH, USA). Subject matter: UQFF structural resolution of the Loschmidt reversibility paradox via $F_{TRZ} \times \beta_i = 0.0603$ CW/CCW chirality imbalance at the SCm/UA vacuum manifold, with $K_{MEX} \times F_{TRZ} = 0.208$ entropy-production rate as integer-primitive identity, requiring no low-entropy past hypothesis.